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# SSM Optimization

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Weekly Meeting

13/02/26

# bitsandbytes

- After some debugging, I have seen that the high perplexity values were due to multi-GPU execution

- If models (fp16 and int{4,8}) are on the same GPU:

```
"dataset": "wikitext2", "perplexity": 21.04495620727539, "kl_divergence": 0.008312617262748822
```

- If models (fp16 and int{4,8}) are executed on two different GPUs:

```
"dataset": "wikitext2", "perplexity": 142726.53125, "kl_divergence": 0.0,
```

- Tried also with transformers==4.53.0 instead of transformers==5.3.0, but it yields the same issue

# Comparison with baselines

## C4

| C4         |                                                                                              |            |               |            |               |              |               |             |               |
|------------|----------------------------------------------------------------------------------------------|------------|---------------|------------|---------------|--------------|---------------|-------------|---------------|
| Model      | Quantized                                                                                    | SINQ       |               | RTN        |               | bitsandbytes |               | HQQ         |               |
|            |                                                                                              | Perplexity | KL divergence | Perplexity | KL divergence | Perplexity   | KL divergence | Perplexity  | KL divergence |
| Mamba-1.4B |             | 14.724     | -             | -          | -             | -            | -             | -           | -             |
| Mamba-1.4B |  (8 bits)   | 14.724     | 0.000130      |            |               |              |               | 14.725      | 0.000125      |
| Mamba-1.4B |  (4 bits)   | 15.117     | 0.0276        |            |               | 15.829       | 0.0751        | 15.199      | 0.033         |
| Mamba-1.4B |  (3 bits)   | 16.904     | 0.143         |            |               | -            | -             | 17.289      | 0.164         |
| Mamba-1.4B |  (2 bits)   | 65.326     | 1.512         |            |               | -            | -             | 74.724      | 1.635         |
| Qwen3-1.7B |             | 19.215     | -             | -          | -             | -            | -             | -           | -             |
| Qwen3-1.7B |  (8 bits) | 19.209     | 0.000359      | 19.229     | 0.000898      | 19.130       | 0.00892       | 19.219      | 0.000865      |
| Qwen3-1.7B |  (4 bits) | 19.870     | 0.073         | 20.814     | 0.125         | 23.442       | 0.219         | 21.332      | 0.142         |
| Qwen3-1.7B |  (3 bits) | 25.245     | 0.387         | 31.153     | 17.787        | -            | -             | 195.840     | 2.222         |
| Qwen3-1.7B |  (2 bits) | 20077.678  | 7.325         | 229633.141 | 10.041        | -            | -             | 1072167.625 | 11.572        |

# Comparison with baselines

## C4

| C4         |            |            |               |            |               |              |               |            |               |
|------------|------------|------------|---------------|------------|---------------|--------------|---------------|------------|---------------|
| Model      | Quantized  | SINQ       |               | RTN        |               | bitsandbytes |               | HQQ        |               |
|            |            | Perplexity | KL divergence | Perplexity | KL divergence | Perplexity   | KL divergence | Perplexity | KL divergence |
| Mamba-2.8B | 🚫          | 13.269     | -             | -          | -             | -            | -             | -          | -             |
| Mamba-2.8B | ✅ (8 bits) | 13.270     | 0.000141      |            |               |              |               | 13.270     | 0.000128      |
| Mamba-2.8B | ✅ (4 bits) | 13.575     | 0.0254        |            |               | 13.886       | 0.0521        | 13.557     | 0.0250        |
| Mamba-2.8B | ✅ (3 bits) | 15.019     | 0.132         |            |               | -            | -             | 14.929     | 0.127         |
| Mamba-2.8B | ✅ (2 bits) | 76.949     | 1.791         |            |               | -            | -             | 51.460     | 1.386         |
| Qwen3-4B   | 🚫          | 16.626     | -             | -          | -             | -            | -             | -          | -             |
| Qwen3-4B   | ✅ (8 bits) | 16.621     | 0.000268      | 16.629     | 11.071        | 16.645       | 0.009         | 16.633     | 0.000313      |
| Qwen3-4B   | ✅ (4 bits) | 17.482     | 0.0639        | 17.541     | 10.745        | 19.296       | 0.141         | 17.501     | 0.074         |
| Qwen3-4B   | ✅ (3 bits) | 21.080     | 0.330         | 21.478     | NaN           | -            | -             | 23.277     | 0.412         |
| Qwen3-4B   | ✅ (2 bits) | 59579.773  | 8.596         | 67191.055  | 12.241        | -            | -             | 41901.063  | 8.347         |

Also on this benchmark SSMs seem to be quantizable with standard LLM techniques, appearing even more robust than LLMs of similar size at low bit-widths (2)

# Comparison with baselines

## C4

| ... running ... |                                                                                              |            |               |            |               |              |               |            |               |
|-----------------|----------------------------------------------------------------------------------------------|------------|---------------|------------|---------------|--------------|---------------|------------|---------------|
| C4              |                                                                                              |            |               |            |               |              |               |            |               |
| Model           | Quantized                                                                                    | SINQ       |               | RTN        |               | bitsandbytes |               | HQQ        |               |
|                 |                                                                                              | Perplexity | KL divergence | Perplexity | KL divergence | Perplexity   | KL divergence | Perplexity | KL divergence |
| Falcon Mamba 7B |             | 10.016     | -             | -          | -             | -            | -             | -          | -             |
| Falcon Mamba 7B |  (8 bits)   |            |               |            |               |              |               |            |               |
| Falcon Mamba 7B |  (4 bits)   |            |               |            |               | 10.368       | 0.0411        |            |               |
| Falcon Mamba 7B |  (3 bits)   |            |               | 11.045     | NaN           | -            | -             |            |               |
| Falcon Mamba 7B |  (2 bits)   |            |               | 52.903     | NaN           | -            | -             |            |               |
| Qwen3-8B        |             | 13.292     | -             | -          | -             | -            | -             | -          | -             |
| Qwen3-8B        |  (8 bits) |            |               |            |               |              |               |            |               |
| Qwen3-8B        |  (4 bits) |            |               |            |               |              |               |            |               |
| Qwen3-8B        |  (3 bits) |            |               |            |               | -            | -             |            |               |
| Qwen3-8B        |  (2 bits) |            |               |            |               | -            | -             |            |               |

# Quamba code

- Working on setting up and run their code.
  - Currently libraries and builds issues on the university server.
  - Switched to the local workstation here and I managed to install all the required libraries.
- Managed to make inferences on their w4a8 checkpoint
- Next: evaluation on Wikitext2

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# Next steps

- Evaluation on Wikitext2/C4 with Quamba-quantized models (check if there are checkpoints for Falcon Mamba-7B)
  - Where?
- Define next direction of work